

PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

Anonymous ACL submission

Abstract

Enterprise property graphs vary widely in schema structure, internal terminology, domain assumptions, governance constraints, and user interaction patterns. A deployment-relevant Text2Cypher benchmark therefore reflects the questions users and agents actually ask of that graph. Creating such a benchmark is difficult because schemas and values are unique, and graph structure changes over time. Each NL-query pair must also be executable, use real graph entities, preserve diversity, and remain balanced across query types and difficulty levels. We present PIPE-Cypher, a local benchmark-generation pipeline that turns a live property graph and optional seed queries from customer questions, analyst logs, or agent tool calls into balanced NL-to-Cypher benchmarks. PIPE-Cypher combines schema profiling, reverse-query grounding, constrained generation, deterministic Cypher governance, execution validation, redaction, diversity controls, and a calibrated local LLM judge. Using local Qwen3.5-9B generation and judging, PIPE-Cypher exports 3,000 accepted FinBench/SNB examples, completes three audited ablation suites, calibrates judge behavior with human labels, and evaluates 11 local downstream models. The resulting benchmark is deliberately discriminative: zero-shot transfer is weak, while a few-shot control shows that schema-specific example banks can help compatible model families. Together, PIPE-Cypher makes Text2Cypher benchmarking a repeatable process that evolves with the graph, its users, and its target workloads.

1 Introduction

Property graphs are attractive in enterprise settings because the facts of interest are often relational paths: account transfers, identity entitlements, access chains, customer interactions, supplier dependencies, and fraud rings. Cypher gives analysts a compact language for these patterns. As LLMs be-

come natural-language interfaces for graph analytics, an organization cannot evaluate a Text2Cypher system only on public schemas; it needs to know whether the model handles its labels, relationship directions, values, governance rules, and recurring operational questions.

For evaluation, that privacy boundary matters as much as model accuracy. A static public dataset is useful for shared comparison, but it cannot contain a bank’s account taxonomy, an identity team’s permission graph, or the categorical values that make a query answerable. It also cannot change when the production schema changes. Industry teams therefore need something different: a repeatable way to turn a live property graph into a balanced, executable, privacy-aware NL-to-Cypher benchmark without sending sensitive schema or values to paid generation APIs.

One might ask why a general AI agent cannot build this benchmark directly. For a small one-off study, a strong model with tools can inspect a schema, draft Cypher, execute it, and revise. Enterprise refreshes recur as schemas and workflows change, often with smaller local models chosen for cost, latency, and privacy. PIPE-Cypher constrains the loop: the model writes questions and Cypher, while graph checks ground values, preserve directions, reject unsafe or ambiguous examples, balance coverage, and leave an audit trail.

PIPE-Cypher addresses this setting. The pipeline profiles a target graph, finds graph values that make candidate questions answerable, constrains a local generator, validates and repairs the generated Cypher, executes it, and then asks a local LLM judge to review only candidates that already have execution evidence. We treat Cypher correctness as something to check, not something to hope the prompt induces: relationship direction, read-only safety, exact literal use, categorical values, contextual return columns, and `RETURN DISTINCT` are enforced before examples are accepted.

Contributions. We make four contributions: (1) a local-model workflow for generating private NL-to-Cypher benchmarks from an organization’s own graph; (2) outcome-aware reverse grounding and Cypher-specific validators for read-only safety, relationship direction, exact literals, categorical values, contextual returns, and conservative rewrites; (3) a scaled public-proxy evaluation over FinBench, SNB, and ICIJ with ablations, judge calibration, redaction audits, and an 11-model local transfer study; and (4) reproducibility artifacts for onboarding, value sampling, benchmark refresh, evidence packaging, and appendix-level audit.

2 Related Work

Recent Text2Cypher resources, including Mind the Query (Chauhan et al., 2025), SyntheT2C (Zhong et al., 2025), Auto-Cypher (Tiwari et al., 2025), Text2Cypher (Ozsoy et al., 2025), CypherBench (Feng et al., 2025), and the public Text2Cypher-2024 corpus (Neo4j, 2024), show that good Cypher data needs schema grounding, execution checks, verification, and complexity-aware evaluation. PIPE-RDF (Ranganath, 2026) makes a related benchmark-factory argument for RDF/SPARQL, using reverse querying, category-balanced generation, retrieval, deduplication, execution validation, and deployment metrics. For enterprise Cypher benchmark generation, PIPE-Cypher adds outcome-aware reverse grounding before natural-language realization, deterministic property-graph governance before export, local judge calibration after execution, explicit privacy and value policies, and provenance-rich refresh artifacts for organizations that need to benchmark their own graphs.

Mind the Query is especially relevant because it reports an Industry Track Text2Cypher dataset with schema, runtime, value, and human logical validation. PIPE-Cypher keeps that validation discipline but changes the object of study. We are not proposing one more static dataset or tuning corpus. We are proposing the process an organization would use to generate, refresh, audit, and redact a benchmark for its own graph, local model endpoint, and value policy.

Text-to-SQL benchmarks such as Spider 2.0 (Lei et al., 2024) and BIRD (Li et al., 2023) have pushed text-to-query evaluation toward realistic database tasks and execution-based scoring, while execution-guided decoding shows why query ex-

ecution is useful semantic feedback rather than a cosmetic check (Wang et al., 2018). Recent residual-skill Text-to-SQL work further shows that optimizing complementary agent skills on residual failures can improve selected accuracy across SQL dialects (Zhu et al., 2026). Graph query generation adds different failure modes: relationship direction can invert the meaning of a query, node and relationship properties live in different namespaces, and path-shaped questions can be syntactically valid while semantically ungrounded. CIKM Auto-Query (Zheng et al., 2024) motivates treating workload generation as a separate object of study from downstream model quality. LDBC FinBench (Qi et al., 2023) and SNB (Püroja et al., 2023) provide public graph workloads with financial and social-network structure, while ICIJ Offshore Leaks gives an additional public finance/compliance onboarding check.

For benchmark quality, lexical diversity alone is not enough. A question bank can use varied wording and still overuse the same graph values or the same Cypher template. PIPE-Cypher therefore combines text-generation diagnostics such as Distinct-n (Li et al., 2016) and self-BLEU-style redundancy checks (Zhu et al., 2018) with text-to-query structure metrics: schema coverage, relationship/property coverage, query-signature diversity, structural feature rates, and normalized entropy over graph/category/difficulty cells. For subset construction, we use an MMR-style novelty objective (Carbonell and Goldstein, 1998) over Cypher signatures, template families, structural substructures, schema atoms, values, and question tokens.

3 Method

PIPE-Cypher has six stages: schema profiling, workload planning, reverse Cypher grounding, constrained generation and repair, deterministic validation and execution, and LLM-judge review. The central design choice is simple: accepted examples should prove that they are answerable and safe. Prompts can ask a model to respect relationship directions or exact literals, but accepted examples must pass schema checks, parser-style structure extraction, live execution, and judge review.

Schema profiling records labels, relationship types, properties, observed directions, and bounded low-cardinality categorical values. Workload planning targets eight categories that appear in operational graph analytics: simple retrieval, com-

plex retrieval, simple aggregation, complex aggregation, boolean existence, negation/difference, path/temporal transaction, and ranking/top-k. Reverse grounding then runs read-only Cypher to find slot values that actually produce rows. This step avoids a common synthetic-data failure: a plausible-looking question that has no answer in the graph.

The deterministic layer checks read-only safety, syntax shape, labels and properties from the schema, explicit relationship types, observed relationship directions, schema-provided categorical values, and non-empty execution where required. Live execution uses read-only credentials and read-access sessions; token-level write rejection is only the first safety check. A lightweight Cypher analyzer extracts return aliases, variables, labels, relationship patterns, risky constructs, and rewrite skip reasons. This makes normalization auditable instead of a silent string edit. The direction gate reads both outgoing and incoming arrow syntax before schema checking, and rejects undirected relationship patterns when direction is required.

4 Implementation

PIPE-Cypher is implemented as a Python package around the pieces a private deployment has to control. The graph is accessed through a read-only client. Before any model is called, a profiler records the schema, relationship directions, and value samples that the run is allowed to use. Generation, validation, judging, and export then operate on this graph profile and on execution traces rather than on backend-specific objects. We use Neo4j to run the experiments, but the method is written around Cypher over property graphs; the graph client is the only backend-specific part of the code. Benchmark generation and judge review use a local Qwen3.5-9B endpoint (Qwen, 2026) behind a vLLM/OpenAI-compatible interface. Downstream evaluation uses 11 completed locally served checkpoints from general instruction, code-tuned, Cypher-tuned, and Text2Cypher-tuned families. We keep these roles separate because they answer different questions: one local endpoint builds the benchmark, and the downstream study measures how other local models behave on the exported examples. All generation and evaluation run inside the organization’s compute boundary without paid generation APIs.

The graph profile is where we make graph-

specific assumptions explicit. For FinBench, the profile comes from the public datagen snapshot export and includes typed node and relationship properties. For SNB, it comes from the official Neo4j/Cypher headers and read-query files. Live schema inspection also records bounded low-cardinality string values as categorical constraints for prompting and validation. During FinBench import, we create rather than merge transaction relationships so repeated account-to-account events remain visible to path and aggregation queries. These details are easy to miss, but they decide whether a generated question is actually answerable. A property may be valid on a relationship but not on a node. A value may be safe to sample or too sensitive to expose. A relationship may only make sense in one direction. A company-owned onboarding run produces the same kind of profile before the first LLM call.

For generation, PIPE-Cypher starts from workload templates whose slots are filled by reverse-binding queries. It can also use a mixed mode in which the LLM proposes additional templates after seeing proven workload seeds. Every run records both accepted and rejected candidates. Retrieved few-shot examples replace graph-specific values with typed placeholders, so the model sees the query structure without repeatedly seeing the same tenant values. A lightweight value grounder adds typed annotations for categorical values and reverse-bound entities, including punctuation variants, possessives, plurals, synonyms, name partials, and small typos.

Inspired by Mind the Query’s prompt-setting analysis, PIPE-Cypher exposes prompt profiles for schema-only, instructions-only, examples-only, examples-plus-instructions, and full governed generation. We report a profile only when it passes the same target-size and evidence checks as the main run.

For LLM-judge review, we do not send the entire schema when a query touches only a small part of it. The judge prompt includes the labels, relationship types, and properties mentioned by the candidate query, while deterministic validators still check against the full schema. This keeps local 9B prompts manageable on larger graphs without weakening schema validation.

The Cypher layer uses constraints drawn from production Text2Cypher work: schema-only prompting, exact matching, relationship direction discipline, RETURN DISTINCT, reserved vari-

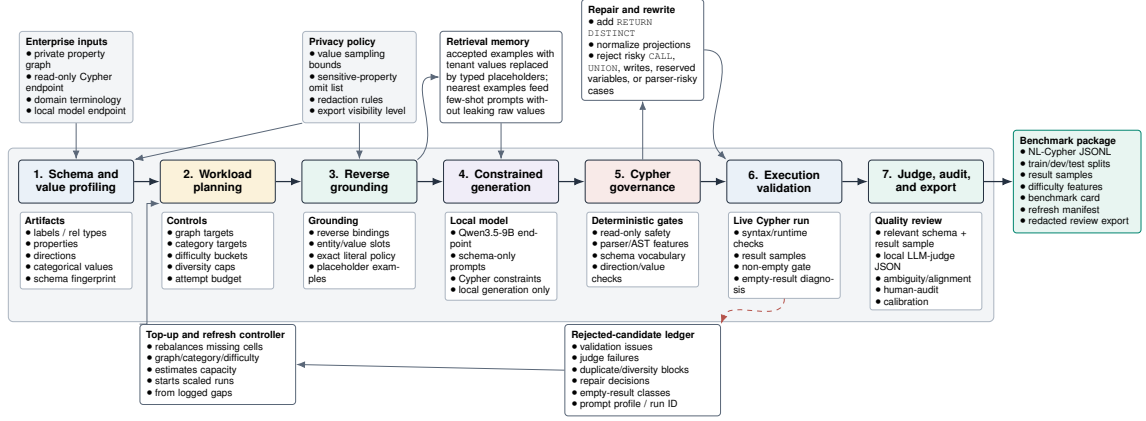


Figure 1: PIPE-Cypher benchmark generation pipeline. The key industry additions beyond static Text2Cypher dataset construction are privacy/value policies, reverse grounding, Cypher governance, execution diagnostics, local judge calibration, and benchmark export/refresh.

Gate	Evidence checked	Failure mode caught
Read-only safety	blocked Cypher write/admin tokens	destructive or operational queries
Schema validity	labels, relationship types, properties, categorical values	hallucinated graph vocabulary or values
Direction validity	observed relationship directions	reversed or invalid traversals
Cypher analysis and normalization	structure extraction, rewrite safety, RETURN DISTINCT	unsafe rewrites or duplicate/noisy rows
Question constraints	quoted values use exact literals	fuzzy matching of exact user values
Execution	query runs and returns rows	syntactic validity without answerability
LLM judge	ambiguity, semantic alignment, schema use	executable but semantically weak examples

Table 1: PIPE-Cypher candidate acceptance gates.

able rejection, categorical values, required contextual return columns, fuzzy value annotations, placeholderized retrieval examples, and parser-aware rewrite boundaries. PIPE-Cypher records parser-style structure features and skips rewrites for risky constructs such as UNION, CALL, UNWIND, WHERE EXISTS, multiple WHERE clauses, or reserved variables.

We also estimate seed-template capacity before full generation. This check caught and fixed a scale blocker: reverse-binding execution was hard-capped to 10 rows, and no-slot negation/ranking seeds could not support full category targets. PIPE-Cypher now uses the configured binding limit and includes slotted negation and ranking seeds for both graphs.

For enterprise onboarding, PIPE-Cypher includes a deployment template for a company’s own graph: read-only credentials, local model endpoint, schema introspection, privacy policy, dry run, scaled run, audit, and redacted export. We validate this pattern on three public proxy graphs rather than a proprietary tenant deployment. Configurable value-sampling policies decide which

Graph	Target/category	Binding limit	Min seed capacity	All categories meet target
FinBench	250	300	300	Yes
SNB	125	200	200	Yes

Table 2: Built-in seed-template capacity after removing the reverse-binding 10-row cap and adding slotted negation/ranking seeds. Capacity is a theoretical scale-readiness check; final examples still must pass validation, execution, diversity, and judge gates.

low-cardinality graph values may enter prompts. Redacted exports replace quoted literals, entity values, and string-valued result samples with stable placeholders for broader internal review. To support schemas beyond the built-in LDBC profiles, PIPE-Cypher derives relationship-count, anti-join, and top-k templates from observed labels, relationship directions, and safe low-cardinality properties, then grounds slot values with outcome-aware reverse Cypher.

5 Experiments

Research questions. We evaluate four questions that matter for an industry benchmark generator: RQ1, can a local-model pipeline produce a balanced executable benchmark over live property

Graph	Candidates	Accepted	Acceptance	Categories at target
FinBench	3,405	2,000	0.587	8/8
SNB	1,520	1,000	0.658	8/8
Total	4,925	3,000	0.609	16/16

Table 3: Full live generation with local Qwen3.5-9B. Candidate counts include the initial sequential run and category-specific recovery top-ups.

graphs? RQ2, do Cypher-specific validation and grounding steps make generation reliable at scale? RQ3, does the resulting benchmark expose meaningful downstream Text2Cypher failures rather than merely checking syntax? RQ4, can the same workflow onboard a new public enterprise-style graph without hard-coding FinBench or SNB?

The generated benchmark contains 3,000 accepted examples: 2,000 from LDBC FinBench and 1,000 from LDBC SNB, balanced over eight workload categories. We report three completed FinBench/SNB ablation suites: target-50, corrected target-100, and a seed-17 target-50 repeat. Downstream Text2Cypher evaluation uses live execution accuracy and answer-set F1 as primary metrics; reference-based text metrics are supported for debugging only and reported in the appendix. We additionally report ICIJ Offshore Leaks as a third public finance/compliance onboarding proxy.

6 Results

RQ1: executable benchmark generation. The full live run produced 3,000 accepted examples from 4,925 candidates using local Qwen3.5-9B for generation and judging. Category-specific recovery top-ups filled the only under-target categories from the initial sequential run. Every exported example passed read-only, syntax, schema, execution, non-empty result, and judge gates.

The accepted records are exported with stable identifiers, train/dev/test splits, result samples, gate metadata, aggregate statistics, and a manifest hash; the appendix gives the full artifact distribution.

Diversity and residual concentration. We do not reduce diversity to one score. The full export has perfect category balance, near-perfect difficulty balance, 1,115 unique grounded entity values, and exact quotation of grounded values in 82.6% of examples with entity bindings. Query-signature diversity remains low because seeded graph-grounded templates carry much of the generation load. Table 4 shows that diversity is still governable after acceptance: at the same graph/category target, a selector using query signatures, template families,

structural substructures, schema atoms, values, and question tokens improves structural coverage, adjusted Distinct-2, query-signature ratio, and property coverage. The result is a balanced executable benchmark, plus clear diagnostics for what should diversify during refresh.

Metric	Random	Governed
PIPE-Diversity index	0.557	0.575
Unique query-signature ratio	0.062	0.135
Structural substructures	97	134
Adjusted Distinct-2	0.266	0.284
Property coverage	0.407	0.426

Table 4: Diversity-governed selection at the same graph/category target. The selector improves structural, lexical, signature, and schema coverage after all quality gates have already passed; the appendix reports the full diversity audit, including residual template concentration.

RQ2: governed generation. Most full-run rejections come from duplicate/diversity controls or empty execution results. Only 2 of 4,925 candidates were schema-invalid after the Cypher checks. A rewrite audit found that all reported-run candidates were already identical to their normalized Cypher, so accepted examples do not depend on semantics-changing rewrites. The ablations should therefore be read as a reliability study of the execution-grounded core: in the target-100 suite, every non-unconstrained graph/setting cell reached all eight category targets, while unconstrained generation did not produce balanced executable coverage. Across the three evidence-ready suites, target-normalized coverage is 1.000 for every non-unconstrained cell.

Judge calibration. An 80-row post-hoc human audit sampled accepted and rejected candidates across both graphs and all categories. Agreement is 80.0%, Cohen’s $\kappa = 0.60$, judge precision/specificity are 1.00, recall is 0.714, and no false accepts appear in the sample. The judge is conservative and protects accepted-example quality; human labels calibrate the gate but do not participate in generation.

RQ3: downstream stress test. We evaluate downstream Text2Cypher by giving each model the schema text and the question, then scoring the generated query by live execution on FinBench or SNB. This is where the benchmark becomes useful: many outputs parse, mention plausible schema, and still answer the wrong graph question. The local

Qwen3.5-9B baseline, for example, reaches 0.963 parse validity and 0.916 schema validity but only 0.189 exact execution accuracy on the 296-example held-out split. In an 11-model completed local transfer suite, zero-shot execution accuracy ranges from 0.000 to 0.203 (mean 0.036). Table 5 gives the control comparison. The primary few-shot generalization result is the scored no-signature control, which excludes exact query-signature matches and near-duplicate questions and raises mean accuracy to 0.200. Ordered and random same-category example banks reach 0.269/0.267, but we treat them as operational upper bounds because they often share query signatures. This split is important for industry use: the benchmark can be a hard model-evaluation set and, separately, a private example bank for schema-specific retrieval or adaptation.

Condition	Sig.	Mean	Best
Zero-shot	–	0.036	0.203
No-signature	0.000	0.200	0.828
Ordered	0.866	0.269	0.993
Random	0.854	0.267	0.986

Table 5: Eleven-model local downstream transfer controls on the 296-example held-out split. Sig. is the fraction of selected demonstrations sharing the test query signature; the no-signature row is the leakage-aware example-bank result, while ordered and random same-category rows are operational upper bounds.

RQ4: third-graph onboarding. ICIJ onboarding reaches 800 accepted examples from 983 candidates on a 2.0M-node, 3.3M-edge public finance/compliance graph, with 100 examples in every category. This is evidence from a third public graph, not proof of private-tenant coverage. It is still important because it exercises schema-derived relationship-count, anti-join, and top-k templates beyond the two LDBC workloads.

7 Industry Use

Enterprise graphs are usually specialized artifacts, not generic benchmark schemas. They encode a company’s products, risk rules, permissions, data integrations, and analyst vocabulary. A Text2Cypher system is useful only if it works on the questions users actually ask of that graph. PIPE-Cypher treats seed queries as a practical bridge from deployment to evaluation: when available, they can come from historical customer questions, analyst query logs, or agent tool calls triggered by user requests. The pipeline then expands those

seeds into a balanced benchmark that tests the same kinds of operations the organization expects its agent to perform.

This matters because the downstream stress test in Table 5 and Appendix Figure 11 shows that general Text2Cypher models often do not transfer cleanly to a new graph with its own schema. The generated examples are therefore useful in two ways. First, they form a private held-out test set for measuring agent behavior under the organization’s schema, values, and safety rules. Second, accepted examples can become a schema-specific question–query bank for retrieval-augmented prompting; the no-signature few-shot control improves mean local-model accuracy, while same-category banks give an operational upper bound (Appendix Tables 18–20). For more complex deployments, the same accepted pairs can also seed supervised adaptation, although we do not claim a tenant-specific fine-tuning result here.

PIPE-Cypher is meant to be rerun when an enterprise graph changes. Schemas change as teams add products, ingest new sources, or encode new business logic, and static Text2Cypher corpora become stale quickly. Each example records the schema snapshot, graph profile, model identifier, validation gates, execution sample, judge scores, difficulty features, and source run, so the benchmark can be refreshed and audited. A deployment needs read-only graph credentials, schema introspection, a bounded value policy, a local endpoint, a dry run, scaled generation, judge calibration, and redacted export. Local inference keeps generation inside the compute boundary; if an organization later permits remote inference, the same value-sampling and redaction policies provide a safer artifact boundary. We validate this pattern on FinBench, SNB, and ICIJ while keeping the private-tenant gap explicit.

8 Conclusion

PIPE-Cypher reframes Text2Cypher benchmarking as a private, repeatable enterprise workflow. The main lesson is that generation improves when Cypher constraints become executable checks. Reverse grounding makes questions answerable. Deterministic validation catches unsafe or schema-invalid queries. Execution exposes empty or brittle candidates. Diversity diagnostics reveal concentration. A calibrated local judge adds a conservative semantic filter. Together these pieces produce a benchmark that is balanced, auditable, refreshable,

and able to reveal downstream model failures that syntax-only evaluation would hide.

Limitations

Execution validity does not guarantee semantic correctness. The completed 80-row, single-human-annotator calibration suggests a conservative judge with no observed false accepts in the labeled sample, but the confidence interval is wider than the point estimate and larger multi-annotator audits may reveal additional failure modes. FinBench, SNB, and ICIJ are public enterprise-style proxies rather than a proprietary tenant graph, so we test the onboarding pattern but not every deployment constraint of a real organization. The full export is balanced by graph, category, and difficulty, but query-signature diagnostics still show template concentration from seeded, execution-grounded generation. PIPE-Cypher deliberately disallows or skips risky Cypher constructs such as writes, undirected relationships, UNION, CALL, UNWIND, and parser-risky rewrites. This is a safe benchmark-generation subset, not complete coverage of every production Cypher idiom. The downstream few-shot result should be read as graph-specific example-bank conditioning: the no-signature control is the leakage-aware result, while ordered and random same-category demonstrations are upper-bound conditions that often share query signatures. Tenant-specific fine-tuning remains an engineering path enabled by the artifact, not a completed deployment claim here. The redaction audit checks exact residuals for known value-bearing strings, but it is not a full PII classifier and does not make schema names confidential.

Ethics Statement

PIPE-Cypher is designed for private benchmark generation under local-model deployment constraints. Organizations using it should restrict benchmark artifacts to authorized users, review sampled values for sensitive content, and document whether judge calibration relied on human annotators. For this study, one external human annotator labeled an 80-row post-hoc judge-calibration packet. The annotator was informed that the labels would be used for research calibration and paper reporting, and raw value-bearing annotation rows are not released. The protocol received an IRB exemption; identifying review-board details are omitted for double-blind submission. Human labels were

never used as a generation gate.

References

- aigentx. 2025a. [llama-3.1-8b-instruct-cypher](#). Hugging Face model repository. Checkpoint source. Accessed: 2026-06-06.
- aigentx. 2025b. [llama-3.1-8b-instruct-cypher-mixed-samples](#). Hugging Face model repository. Checkpoint source. Accessed: 2026-06-06.
- Azzedde. 2025. [llama3.1-8b-text2cypher](#). Hugging Face model repository. Checkpoint source. Accessed: 2026-06-06.
- Satanjeev Banerjee and Alon Lavie. 2005. [METEOR: An automatic metric for MT evaluation with improved correlation with human judgments](#). In *Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization*, pages 65–72, Ann Arbor, Michigan. Association for Computational Linguistics.
- Jaime Carbonell and Jade Goldstein. 1998. [The use of MMR, diversity-based reranking for reordering documents and producing summaries](#). In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 335–336. ACM.
- Vashu Chauhan, Shobhit Raj, Shashank Mujumdar, Avirup Saha, and Anannay Jain. 2025. [Mind the query: A benchmark dataset towards Text2Cypher task](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1890–1905, Suzhou, China. Association for Computational Linguistics.
- Yanlin Feng, Simone Papicchio, and Sajjadur Rahman. 2025. [CypherBench: Towards precise retrieval over full-scale modern knowledge graphs in the LLM era](#). *Preprint*, arXiv:2412.18702.
- Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, Louis Rouillard, Thomas Mesnard, and 1 others. 2025. [Gemma 3 technical report](#). *Preprint*, arXiv:2503.19786.
- Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, Johan Ferret, Peter Liu, and 1 others. 2024. [Gemma 2: Improving open language models at a practical size](#). *Preprint*, arXiv:2408.00118.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, and 1 others. 2024. [The Llama 3 herd of models](#). *Preprint*, arXiv:2407.21783.

- Binyuan Hui, Jian Yang, Zeyu Cui, Jiayi Yang, Dayiheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang, Bowen Yu, Keming Lu, Kai Dang, Yang Fan, Yichang Zhang, An Yang, Rui Men, Fei Huang, Bo Zheng, Yibo Miao, Shanghaoran Quan, and 5 others. 2024. [Qwen2.5-coder technical report](#). *Preprint*, arXiv:2409.12186.
- Matthew A. Jaro. 1989. [Advances in record-linkage methodology as applied to matching the 1985 census of tampa, florida](#). *Journal of the American Statistical Association*, 84(406):414–420.
- Moussa Kamal Eddine, Guokan Shang, Antoine Tixier, and Michalis Vazirgiannis. 2022. [FrugalScore: Learning cheaper, lighter and faster evaluation metrics for automatic text generation](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1305–1318, Dublin, Ireland. Association for Computational Linguistics.
- Fangyu Lei, Jixuan Chen, Yuxiao Ye, Ruisheng Cao, Dongchan Shin, Hongjin Su, Zhaoqing Suo, Hongcheng Gao, Wenjing Hu, Pengcheng Yin, Victor Zhong, Caiming Xiong, Ruoxi Sun, Qian Liu, Sida Wang, and Tao Yu. 2024. [Spider 2.0: Evaluating language models on real-world enterprise text-to-SQL workflows](#). *Preprint*, arXiv:2411.07763.
- Jinyang Li, Binyuan Hui, Ge Qu, Jiayi Yang, Binhua Li, Bowen Li, Bailin Wang, Bowen Qin, Rongyu Cao, Ruiying Geng, Nan Huo, Xuanhe Zhou, Chenhao Ma, Guoliang Li, Kevin C. C. Chang, Fei Huang, Reynold Cheng, and Yongbin Li. 2023. [Can LLM already serve as a database interface? a BIG bench for large-scale database grounded text-to-SQLs](#). In *Advances in Neural Information Processing Systems*.
- Jiwei Li, Michel Galley, Chris Brockett, Jianfeng Gao, and Bill Dolan. 2016. [A diversity-promoting objective function for neural conversation models](#). In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 110–119, San Diego, California. Association for Computational Linguistics.
- Chin-Yew Lin. 2004. [ROUGE: A package for automatic evaluation of summaries](#). In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain. Association for Computational Linguistics.
- Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. *Introduction to Information Retrieval*. Cambridge University Press.
- Neo4j. 2024. [text2cypher-2024v1: Consolidated text-to-Cypher dataset](#). Hugging Face dataset.
- Neo4j. 2025a. [text-to-cypher-Gemma-3-4B-Instruct-2025.04.0](#). Hugging Face model repository. Checkpoint source. Accessed: 2026-06-06.
- Neo4j. 2025b. [text2cypher-gemma-2-9b-it-finetuned-2024v1](#). Hugging Face model repository. Checkpoint source. Accessed: 2026-06-06.
- Makbule Gulcin Ozsoy, Leila Messallem, Jon Besga, and Gianandrea Minneci. 2025. [Text2Cypher: Bridging natural language and graph databases](#). In *Proceedings of the Workshop on Generative AI and Knowledge Graphs*, pages 100–108, Abu Dhabi, UAE. International Committee on Computational Linguistics.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. [Bleu: a method for automatic evaluation of machine translation](#). In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, Philadelphia, Pennsylvania, USA. Association for Computational Linguistics.
- projectwilsen. 2024. [llama3.1-8b-text2cypher-neo4j-live](#). Hugging Face model repository. Checkpoint source. Accessed: 2026-06-06.
- David Püroja, Jack Waudby, Peter Boncz, and Gábor Szárnyas. 2023. [The LDBC social network benchmark interactive workload v2: A transactional graph query benchmark with deep delete operations](#). *Preprint*, arXiv:2307.04820.
- Shipeng Qi, Heng Lin, Zhihui Guo, Gábor Szárnyas, Bing Tong, Yan Zhou, Bin Yang, Jiansong Zhang, Zheng Wang, Youren Shen, Changyuan Wang, Parviz Peiravi, Henry Gabb, and Ben Steer. 2023. [The LDBC financial benchmark](#). *Preprint*, arXiv:2306.15975.
- Qwen. 2026. [Qwen3.5-9B](#). Hugging Face model repository. Checkpoint source. Accessed: 2026-06-06.
- Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. 2016. [SQuAD: 100,000+ questions for machine comprehension of text](#). In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 2383–2392, Austin, Texas. Association for Computational Linguistics.
- Suraj Ranganath. 2026. [PIPE-RDF: An LLM-assisted pipeline for enterprise RDF benchmarking](#). *Preprint*, arXiv:2602.18497.
- Saiprasanth15. 2024. [llama3.1-8b-text2cypher-neo4j-live](#). Hugging Face model repository. Checkpoint source. Accessed: 2026-06-06.
- Aman Tiwari, Shiva Krishna Reddy Malay, Vikas Yadav, Masoud Hashemi, and Sathwik Tejaswi Madhusudhan. 2025. [Auto-cypher: Improving LLMs on cypher generation via LLM-supervised generation-verification framework](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2: Short Papers*, pages 623–640, Albuquerque, New Mexico. Association for Computational Linguistics.
- tomasonjo. 2024. [text2cypher-demo-16bit](#). Hugging Face model repository. Checkpoint source. Accessed: 2026-06-06.

- Chenglong Wang, Kedar Tatwawadi, Marc Brockschmidt, Po-Sen Huang, Yi Mao, Oleksandr Polozov, and Rishabh Singh. 2018. [Robust text-to-SQL generation with execution-guided decoding](#). *Preprint*, arXiv:1807.03100.
- William E. Winkler. 1990. [String comparator metrics and enhanced decision rules in the Fellegi-Sunter model of record linkage](#). In *Proceedings of the Section on Survey Research Methods*, pages 354–359. American Statistical Association.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, and 1 others. 2025. [Qwen3 technical report](#). *Preprint*, arXiv:2505.09388.
- Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. [BERTScore: Evaluating text generation with BERT](#). In *International Conference on Learning Representations*.
- Xiuwen Zheng, Arun Kumar, and Amarnath Gupta. 2024. [Generating cross-model analytics workloads using LLMs](#). In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, pages 4303–4307. ACM.
- Zijie Zhong, Linqing Zhong, Zhaoze Sun, Qingyun Jin, Zengchang Qin, and Xiaofan Zhang. 2025. [SyntheT2C: Generating synthetic data for fine-tuning large language models on the Text2Cypher task](#). In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 672–692, Abu Dhabi, UAE. Association for Computational Linguistics.
- Jiongli Zhu, Haoquan Guan, Parjanya Prajakta Prashant, Nikki Lijing Kuang, Seyedeh Baharan Khatami, Canwen Xu, Xiaodong Yu, Yingyu Lin, Zhewei Yao, Yuxiong He, and Babak Salimi. 2026. [Residual skill optimization for text-to-sql ensembles](#). *Preprint*, arXiv:2605.21792.
- Yaoming Zhu, Sidi Lu, Lei Zheng, Jiaxian Guo, Weinan Zhang, Jun Wang, and Yong Yu. 2018. [Txygen: A benchmarking platform for text generation models](#). In *The 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1097–1100. ACM.

A Prior-Work Positioning

Table 6 expands the related-work comparison from the main paper. PIPE-Cypher’s contribution is the way answerability, governance, privacy, judging, and refresh are assembled into an organization-run benchmark factory.

Prior mechanism	PIPE-Cypher delta
Template/LLM synthesis in SyntheT2C (Zhong et al., 2025) and Auto-Cypher (Tiwari et al., 2025)	Outcome-aware reverse grounding binds slots through live Cypher before NL realization.
Execution validation in Auto-Cypher (Tiwari et al., 2025) and Mind the Query (Chauhan et al., 2025)	Execution is one gate in a ledger with direction, read-only, value, non-empty, and judge evidence.
Schema/value rechecks in Mind the Query (Chauhan et al., 2025)	Checks are packaged for private refresh with configurable value sampling and redacted exports.
PIPE-RDF enterprise benchmark factory (Ranganath, 2026)	Adapts the factory framing from RDF/SPARQL to property graphs, where direction, relationship properties, and path semantics require Cypher-specific governance.
Human logical review in Mind the Query (Chauhan et al., 2025)	Human labels calibrate a local LLM judge; they are not a generation gate.
Public static datasets and tuning corpora (Ozsoy et al., 2025; Feng et al., 2025; Neo4j, 2024)	Organization-run benchmark factory with manifests, refresh, and provenance audits.

Table 6: Prior mechanism comparison. PIPE-Cypher combines outcome-aware grounding, deterministic Cypher checks, local judge calibration, privacy policy, refresh support, and audit logs so organizations can generate their own benchmarks rather than only consume static datasets.

B Experimental Setup and Exported Benchmark

Unless noted otherwise, the extended results use the same live FinBench/SNB export as the main results. Tables in this section pin down the scale, graph mix, split, and validation totals so the downstream and diversity analyses are tied to a single benchmark artifact.

B.1 Benchmark Artifact Summary

Tables 7, 8, and 9 give the run configuration, export manifest summary, and gate/distribution details for the 3,000-example benchmark.

Setting	Value
Accepted examples	3,000
Primary graph	LDBC FinBench, 2,000 examples
Secondary graph	LDBC SNB, 1,000 examples
Categories	8 balanced categories, 375 each
Generation/judge model	Local Qwen3.5-9B
Execution backend	Neo4j Community, two live databases
Judge audit packet	80 sampled accepted/rejected pairs

Table 7: Full live experimental setup used for the generated benchmark artifact.

Export artifact	Examples	FinBench	SNB	Train	Dev	Test
Live full benchmark	3,000	2,000	1,000	2,408	296	296

Table 8: Accepted live full benchmark package with stable IDs, gate metadata, result samples, statistics, and manifest hash `cf274344be2abe7e`.

Artifact property	Value
Categories	8 balanced categories
FinBench/category	250
SNB/category	125
Difficulty split	1,569 easy / 1,431 medium
Used labels / rel. types	16 / 17
Read/syntax/schema/exec/judge gates	3,000/3,000

Table 9: Distribution and gate summary for the exported full benchmark artifact.

C Governed Generation Evidence

The central reliability question is whether reverse grounding and Cypher validation can fill every planned graph/category cell at target scale.

C.1 Target-100 Stress Baseline

Figure 2 and Table 10 show the target-100 stress baseline. The unconstrained rows are deliberately harsh: raw local-model generations may look plausible, but they do not produce balanced, executable benchmark coverage. The governed variants fill every target cell on both graphs.

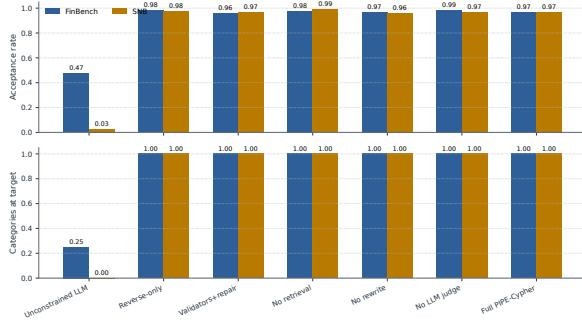


Figure 2: Target-100 ablation yield on FinBench and SNB. Unconstrained local generation is reported as a stress baseline with explicit attempt accounting; the execution-grounded governed variants reach all eight workload-category targets on both graphs. This shows that the grounded governance core is reliable at filling the planned benchmark cells; it does not claim that each optional stage independently increases yield.

Setting	Graph	Attempts	Records	Accepted	Acceptance	Categories at target
Unconstrained LLM	FinBench	422	422	200	0.474	2/8
Unconstrained LLM	SNB	2,000	2,000	50	0.025	0/8
Reverse-only	FinBench	815	815	800	0.982	8/8
Reverse-only	SNB	820	820	800	0.976	8/8
Validators+repair	FinBench	833	833	800	0.960	8/8
Validators+repair	SNB	824	824	800	0.971	8/8
No retrieval	FinBench	819	819	800	0.977	8/8
No retrieval	SNB	809	809	800	0.989	8/8
No rewrite	FinBench	826	826	800	0.969	8/8
No rewrite	SNB	834	834	800	0.959	8/8
No LLM judge	FinBench	812	812	800	0.985	8/8
No LLM judge	SNB	828	828	800	0.966	8/8
Full PIPE-Cypher	FinBench	824	824	800	0.971	8/8
Full PIPE-Cypher	SNB	824	824	800	0.971	8/8

Table 10: Live target-100 ablation evidence with local Qwen3.5-9B. Governed graph runs target 100 accepted examples per category; the unconstrained row is a stress baseline reported with explicit attempt accounting.

C.2 Gate-Level Quality

Yield alone is not enough. A benchmark factory must show which checks keep bad examples out. Table 11 and Figure 3 report read-only, syntax, schema, execution, and judge/post-hoc rates for the same target-100 suite. Read-only and syntax checks saturate once governance is active; execution and semantic judging expose the remaining differences.

Setting	Graph	Read-only	Syntax	Schema	Exec.	Judge/post-hoc
Unconstrained LLM	FinBench	1.000	1.000	0.806	0.723	0.557
Unconstrained LLM	SNB	1.000	0.998	0.599	0.478	0.144
Reverse-only	FinBench	1.000	1.000	0.982	0.982	0.982
Reverse-only	SNB	1.000	1.000	0.998	0.998	0.976
Validators+repair	FinBench	1.000	1.000	1.000	1.000	0.960
Validators+repair	SNB	1.000	1.000	0.998	0.998	0.971
No retrieval	FinBench	1.000	1.000	1.000	1.000	0.977
No retrieval	SNB	1.000	1.000	0.998	0.998	0.989
No rewrite	FinBench	1.000	1.000	1.000	1.000	0.969
No rewrite	SNB	1.000	1.000	0.998	0.998	0.959
No LLM judge	FinBench	1.000	1.000	1.000	1.000	0.985
No LLM judge	SNB	1.000	1.000	0.998	0.998	0.966
Full PIPE-Cypher	FinBench	1.000	1.000	1.000	1.000	0.971
Full PIPE-Cypher	SNB	1.000	1.000	0.998	0.998	0.971

Table 11: Quality-gate rates for the live target-100 ablation suite. Rates are computed over all generated records in each graph/setting; for no-judge settings, the judge column is a post-hoc scoring diagnostic.

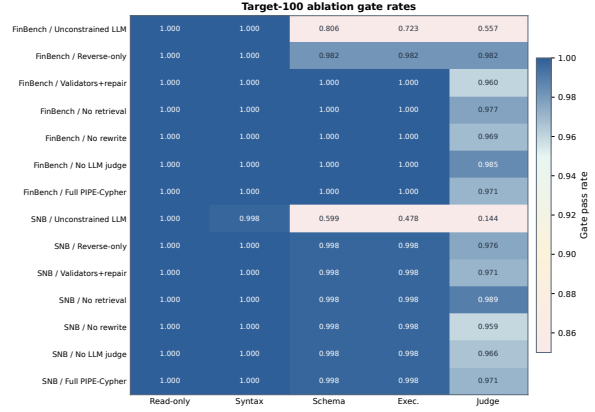


Figure 3: Gate-rate heatmap for the audited target-100 ablation suite. Deterministic read-only and syntax gates are saturated; execution and judge rates expose the remaining quality differences across graph and pipeline variants.

C.3 Repeated-Suite Stability

The same pattern holds across target sizes and seeds. The repeated-suite comparison normalizes by each suite’s planned target, so it measures stability rather than rewarding larger raw counts. Full PIPE-Cypher and the governed ablations reach complete target coverage on both graphs, which is the behavior we want from a repeatable benchmark generator.

Setting	Graph	Suites	Target cov.	Acceptance	Cat. target	Exec.	Judge
No LLM judge	FinBench	3/3	1.000	0.994±0.008	1.000	1.000	0.994
No retrieval	FinBench	3/3	1.000	0.967±0.031	1.000	1.000	0.967
No rewrite	FinBench	3/3	1.000	0.969±0.023	1.000	1.000	0.969
Full PIPE-Cypher	FinBench	3/3	1.000	0.975±0.016	1.000	1.000	0.975
Reverse-only	FinBench	3/3	1.000	0.994±0.011	1.000	0.994	0.994
Validators+repair	FinBench	3/3	1.000	0.987±0.023	1.000	1.000	0.987
No LLM judge	SNB	3/3	1.000	0.982±0.015	1.000	0.996	0.982
No retrieval	SNB	3/3	1.000	0.993±0.004	1.000	0.996	0.993
No rewrite	SNB	3/3	1.000	0.983±0.021	1.000	0.996	0.983
Full PIPE-Cypher	SNB	3/3	1.000	0.987±0.014	1.000	0.996	0.987
Reverse-only	SNB	3/3	1.000	0.989±0.011	1.000	0.996	0.989
Validators+repair	SNB	3/3	1.000	0.987±0.014	1.000	0.996	0.987

Table 12: Target-size and repeated-seed ablation sensitivity. Target coverage normalizes accepted examples by each suite’s planned graph/category target, so target-50 and target-100 suites can be compared without treating larger raw counts as quality gains. Unconstrained local generation is excluded from this stability table and reported separately as the attempt-logged stress baseline in Table 10.

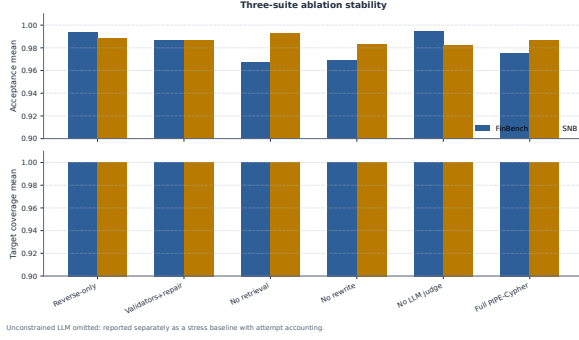


Figure 4: Three-suite ablation stability over the original target-50 suite, corrected target-100 suite, and seed-17 target-50 repeat. Target-normalized coverage stays at 1.000 for all non-unconstrained cells.

C.4 Rejected Candidate Taxonomy

The rejection taxonomy shows where the remaining work occurs. After Cypher validation, schema-invalid candidates are rare. Most rejections are duplicate/diversity blocks from recovery or empty-result candidates caught before export. This is the behavior an enterprise deployment wants: invalid or brittle examples remain in the ledger, not in the benchmark.

Failure bucket	Count	Share of rejected
Diversity/duplicate control	1,366	0.710
Empty result	486	0.252
Judge semantic reject	67	0.035
Execution error	4	0.002
Schema invalid	2	0.001

Table 13: Failure taxonomy over full-run generation candidates before benchmark export. Accepted examples are excluded from the bucket shares.

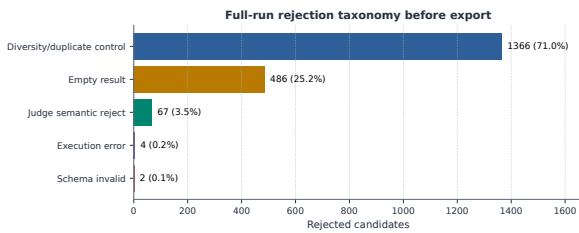


Figure 5: Full-run rejection taxonomy before benchmark export. Most rejected candidates come from duplicate/diversity control during category recovery and from empty execution results; schema-invalid Cypher is rare after Cypher validation.

D Benchmark Artifact and Third-Graph Onboarding

D.1 Export Balance

The exported benchmark makes scale and balance visible. Figure 6 shows the planned 2:1 FinBench/SNB mix, exact category balance, and near-even difficulty split. PIPE-Cypher produces a benchmark that is balanced by design rather than sampled opportunistically.

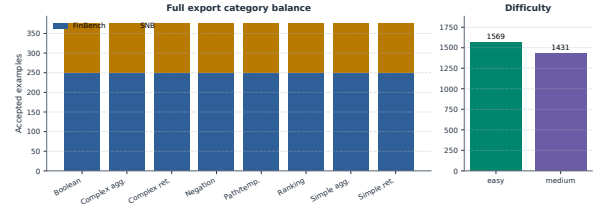


Figure 6: Full 3,000-example export distribution. The benchmark preserves the planned 2:1 FinBench/SNB graph mix while balancing all eight Cypher workload categories and maintaining a near-even easy/medium difficulty split.

D.2 Graph Scale and Schema Variety

Table 14 summarizes the graph sizes and live schema inventories before the third-graph onboarding audit. FinBench and SNB are the controlled LDBC workloads. ICIJ is the public finance/compliance graph we use to test whether the same onboarding code works outside the two original schemas. Figures 7–9 show the same schemas in graph form. Node boxes are labels; arrows are directed relationship families observed by introspection. Some arrows group repeated label-pair alternatives so the figure remains readable, while Table 14 gives the full inventory counts.

Graph	Nodes	Edges	Labels	Rel. types	Patterns	Node props	Rel. props
FinBench	10,006	57,622	5	9	13	37	32
SNB	34,735	70,842	14	15	59	62	5
ICIJ Offshore Leaks	2,016,523	3,339,267	5	14	64	82	48

Table 14: Study graph size and schema inventory from live introspection. Patterns are directed start-label/type/end-label triples; property counts are distinct label-property or relationship-type-property fields. ICIJ Offshore Leaks is used as a public third-graph onboarding audit beyond the two LDBC workloads.

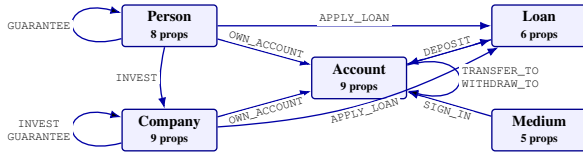


Figure 7: FinBench schema graph used in the reported runs. The workload is centered on financial entities, account ownership, transaction/event relationships, loans, sign-in media, guarantees, and investments.

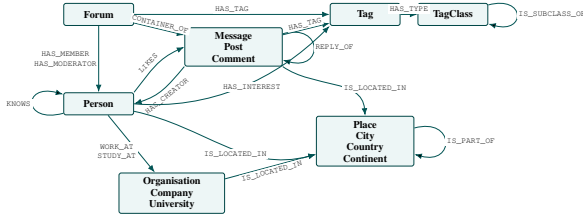


Figure 8: SNB schema graph used in the reported runs. Related labels are grouped into content, place, and organization families so all 14 labels and 59 directed label/type/label patterns remain readable; Table 14 gives the full inventory counts.

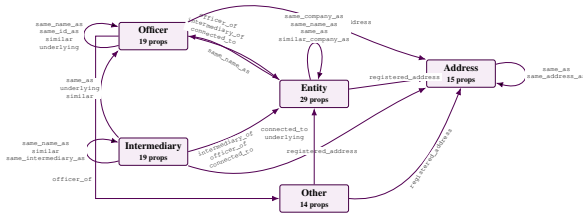


Figure 9: ICIJ Offshore Leaks schema graph used for the third-graph onboarding audit. The schema is compact in node labels but large in edge volume: relationship families encode officer/entity/intermediary roles, registered addresses, identity-resolution links, and similarity links.

D.3 ICIJ Onboarding

Table 15 and Figure 10 report the third-graph result. ICIJ matters because it forced PIPE-Cypher to derive sparse-category templates from the schema instead of relying on preauthored FinBench/SNB templates. The run reaches all eight category targets while using the same validation and audit standards.

ICIJ onboarding property	Value
Graph nodes / relationships	2,016,523 / 3,339,267
Labels / relationship types	5 / 14
Generated / accepted	983 / 800
Acceptance rate	0.814
Categories at target	8/8
Study audit	ready
Sparse schema-derived accepts	complex agg. 97, negation 28, ranking 98

Table 15: ICIJ Offshore Leaks third-graph onboarding audit. The public finance/compliance graph tests arbitrary-schema generation beyond the two LDBC study workloads; raw values remain outside the reported artifacts.

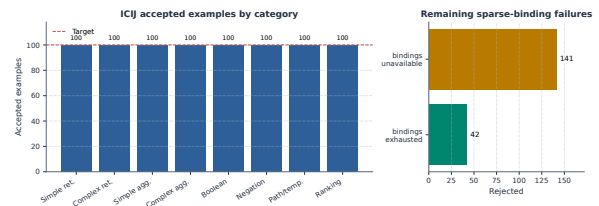


Figure 10: ICIJ Offshore Leaks onboarding audit. Schema-derived sparse-category templates recover balanced target-100 coverage on a public finance/compliance graph beyond the two LDBC workloads.

D.4 Category Crosswalk

Table 16 connects PIPE-Cypher’s eight categories to the simpler retrieval/aggregation/evaluation-query taxonomy used in Mind the Query. The crosswalk keeps the comparison clear while making our extra enterprise workload classes explicit: negation, temporal/path reasoning, and ranking/top-k.

PIPE-Cypher category	Closest MTQ category	Added enterprise distinction
Simple retrieval	SR	Direct node/edge lookup with exact filters.
Complex retrieval	CR	Multi-hop or multi-pattern retrieval.
Simple aggregation	SA	Single aggregation such as count/min/max/average.
Complex aggregation	CA	Grouped or multi-stage aggregation over graph neighborhoods.
Boolean existence	EQ	Precise yes/no or existence answer.
Negation/difference	CR	Absence, anti-join, or difference query.
Path/temporal transaction	CR/CA	Temporal or path-oriented transaction neighborhood.
Ranking/top-k	SA/CA	Ordered top-k query with explicit limit.

Table 16: Category crosswalk to Mind the Query. PIPE-Cypher keeps the familiar retrieval/aggregation/evaluation-query structure while adding enterprise workloads such as negation, temporal paths, and ranking.

E Downstream Transfer and Example-Bank Utility

E.1 Transfer Controls

The downstream experiment tests whether the benchmark is useful for model evaluation. A useful enterprise benchmark should not merely reward syntactically valid Cypher. It should reveal when a model produces a query that parses, mentions plausible schema elements, and even executes,

but answers the wrong operational question. The Qwen3.5-9B result has exactly that shape: parse validity is 0.963 and schema validity is 0.916, while exact execution accuracy is only 0.189 on the full held-out split. The gap shows that PIPE-Cypher measures semantic graph-query competence rather than only query formatting.

The multi-model transfer suite compares local general instruction, code-tuned, Cypher instruction, and Text2Cypher-finetuned models while keeping the schema prompt, evaluation backend, split, and execution metrics fixed. The exact checkpoint and adapter provenance is listed in Table 17. Figure 11 shows a sharp pattern: zero-shot transfer is weak, but accepted graph-specific examples can become a useful private question-answer bank. Demonstration-bank controls close much of the gap for compatible model families such as Qwen, Qwen-Coder, and one Gemma Text2Cypher LoRA. Several public fine-tuned checkpoints remain brittle under enterprise-style prompting.

Table 19 reports checkpoint-level uncertainty for the same controls. The interval is deliberately conservative because the unit is model checkpoint rather than question row; it supports the narrower claim that example-bank gains are model-family dependent, not universal.

Control	Mean acc.	Δ vs zero	95% Δ CI	Models improved
Ordered same-category	0.269	0.233	[0.000, 0.481]	3/11
Scored no-signature	0.200	0.163	[0.000, 0.335]	3/11
Random same-category mean	0.267	0.231	[0.000, 0.464]	3/11

Table 19: Model-level paired bootstrap uncertainty for downstream few-shot controls. The unit of resampling is the local checkpoint, not an individual question, so the interval is a conservative check on whether gains are broad across model families. Zero-shot mean execution accuracy is 0.036.

Table 20 gives the interpretation for the transfer result. Ordered and random same-category demonstrations are useful example-bank conditions. The scored no-signature condition is the stricter leakage-aware control.

Selection mode	Rows	Demos	Sig. match	High sim.	Mean sim.
Ordered	296	1480	0.866	0.389	0.825
No-signature	296	1480	0.000	0.000	0.585
Random	296	1480	0.854	0.409	0.824

Table 20: Few-shot leakage controls for downstream demonstration-bank evaluation. The held-out split has 0 exact train/test question overlaps and 289 train/test query-signature overlaps; selection-mode rates show how often retrieved demonstrations share the test query signature or exceed the 0.90 normalized-question similarity threshold.

Displayed model	Served checkpoint or adapter	Family	Citation
aigentx/Llama-3.1-8B Cypher LoRA	aigentx/llama-3.1-8b-instruct-cypher	Cypher LoRA	(Grattafiori et al., 2024; aigentx, 2025a)
aigentx/Llama-3.1-8B Cypher mixed LoRA	aigentx/llama-3.1-8b-instruct-cypher-mixed-samples	Cypher mixed LoRA	(Grattafiori et al., 2024; aigentx, 2025b)
Azzedde/llama3.1-8b-text2cypher	Azzedde/llama3.1-8b-text2cypher	Text2Cypher fine-tuned	(Grattafiori et al., 2024; Azzedde, 2025)
Gemma-2-9B-IT	google/gemma-2-9b-it	General instruction	(Gemma Team et al., 2024)
neo4j/Gemma-2-9B Text2Cypher LoRA	neo4j/text2cypher-gemma-2-9b-it-finetuned-2024v1	Text2Cypher LoRA	(Gemma Team et al., 2024; Ozsoy et al., 2025; Neo4j, 2025b)
neo4j/Gemma-3-4B Text2Cypher	neo4j/text-to-cypher-Gemma-3-4B-Instruct-2025.04.0	Text2Cypher fine-tuned	(Gemma Team et al., 2025; Ozsoy et al., 2025; Neo4j, 2025a)
projectwilsen/Llama-3.1-8B Text2Cypher LoRA	projectwilsen/llama3.1-8b-text2cypher-neo4j-live	Text2Cypher LoRA	(Grattafiori et al., 2024; projectwilsen, 2024)
Qwen2.5-Coder-7B-Instruct	Qwen/Qwen2.5-Coder-7B-Instruct	Code instruction	(Hui et al., 2024)
Qwen3.5-9B	Qwen/Qwen3.5-9B	General instruction	(Yang et al., 2025; Qwen, 2026)
Saiprasanth15/Llama-3.1-8B Text2Cypher LoRA	Saiprasanth15/llama3.1-8b-text2cypher-neo4j-live	Text2Cypher LoRA	(Grattafiori et al., 2024; Saiprasanth15, 2024)
tomasonjo/text2cypher-demo-16bit	tomasonjo/text2cypher-demo-16bit	Text2Cypher fine-tuned	(Grattafiori et al., 2024; tomasonjo, 2024)

Table 17: Downstream model provenance for the 11 completed local checkpoints in the transfer study. Original model-family papers are cited wherever available; Hugging Face repository citations are retained only to identify exact fine-tuned checkpoints or adapters without a separate paper.

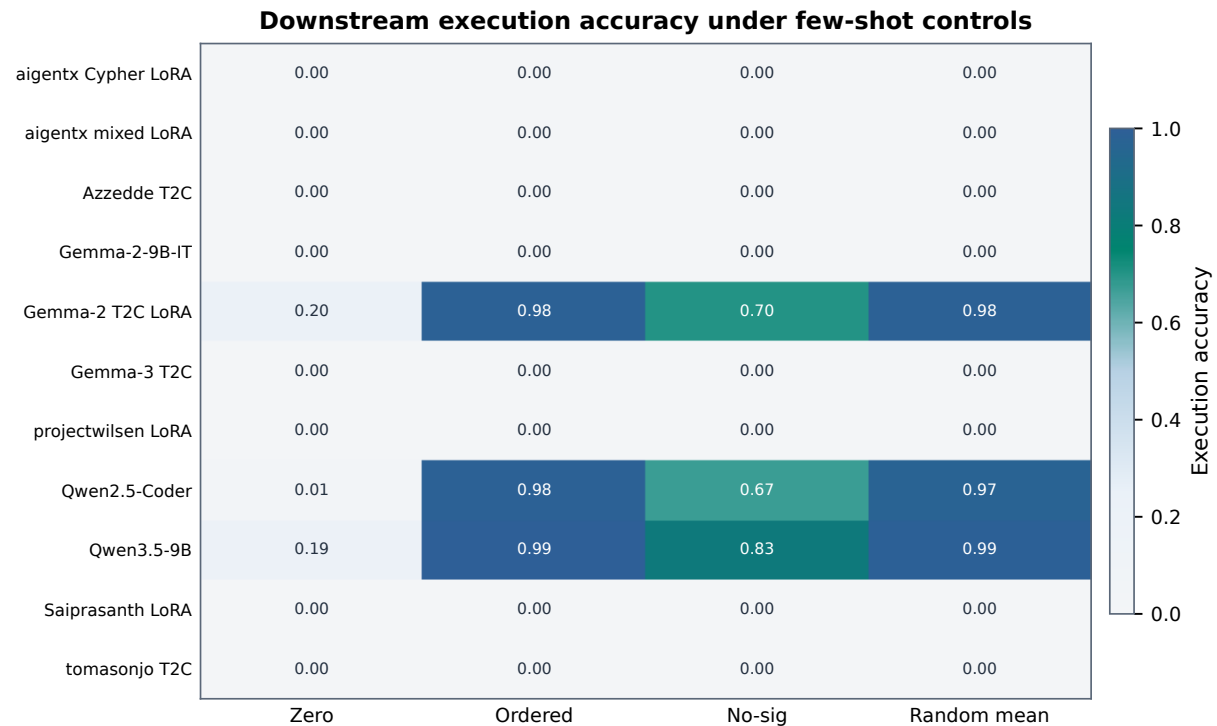


Figure 11: Execution accuracy for 11 completed local downstream models under zero-shot and few-shot control modes. The heatmap highlights both findings: schema-specific examples can sharply help compatible models, and several public fine-tuned checkpoints still fail under enterprise-style Cypher prompting.

Model	Tuning	Zero	Ordered	No-sig	Random μ	Random σ	Best
aigentx/Llama-3.1-8B Cypher LoRA	Cypher LoRA	0.000	0.000	0.000	0.000	0.000	no gain
aigentx/Llama-3.1-8B Cypher mixed LoRA	Cypher mixed LoRA	0.000	0.000	0.000	0.000	0.000	no gain
Azzedde/llama3.1-8b-text2cypher	Text2Cypher fine-tuned	0.000	0.000	0.000	0.000	0.000	no gain
Gemma-2-9B-IT	general instruction	0.000	0.000	0.000	0.000	0.000	no gain
neo4j/Gemma-2-9B Text2Cypher LoRA	Text2Cypher LoRA	0.203	0.983	0.699	0.981	0.009	ordered (0.983)
neo4j/Gemma-3-4B Text2Cypher	Text2Cypher fine-tuned	0.000	0.000	0.000	0.000	0.000	no gain
projectwilsen/Llama-3.1-8B Text2Cypher LoRA	Text2Cypher LoRA	0.000	0.000	0.000	0.000	0.000	no gain
Qwen2.5-Coder-7B-Instruct	code instruction	0.007	0.983	0.669	0.972	0.002	ordered (0.983)
Qwen3.5-9B	general instruction	0.189	0.993	0.828	0.986	0.007	ordered (0.993)
Saiprasanth15/Llama-3.1-8B Text2Cypher LoRA	Text2Cypher LoRA	0.000	0.000	0.000	0.000	0.000	no gain
tomasonjo/text2cypher-demo-16bit	Text2Cypher fine-tuned	0.000	0.000	0.000	0.000	0.000	no gain

Table 18: Few-shot demonstration-bank controls for local downstream Text2Cypher evaluation over completed 296-example local-model runs. Ordered uses the deterministic same-graph, same-category example bank; scored excludes exact query-signature matches and near-duplicate questions; random reports the mean and standard deviation across seeds 13, 17, and 23. “No gain” means no few-shot control exceeded that model’s zero-shot execution accuracy. Model-family papers and exact checkpoint sources are listed in Table 17.

E.2 Downstream Failure Modes

Downstream outcome	Count	Share of incorrect
Answer mismatch	125	0.521
Execution failed	79	0.329
Schema invalid	25	0.104
Parse invalid	11	0.046

Table 21: Downstream Text2Cypher failure taxonomy for local Qwen3.5-9B on the full exported test split. Shares exclude exact-answer matches.

Table 21 makes the downstream study actionable. An answer mismatch means the model produced executable Cypher for the wrong semantics, so better graph-specific examples, retrieval, or adaptation are the likely fixes. Execution failures point instead to unsupported operators, brittle literals, or missing repair rules. Parse and schema failures are the

errors deterministic guards should catch before a query ever reaches users.

This is why the taxonomy appears next to the transfer results. Many incorrect outputs are not malformed queries; they run and return the wrong answer set. A syntax-only or schema-only benchmark would miss that failure. For a benchmark owner, the taxonomy is therefore a debugging interface: it shows whether the next improvement should target schema retrieval, prompt constraints, value grounding, operator repair, or tenant-specific adaptation.

F Diversity and Cypher Strategy Audit

F.1 Diversity Diagnostics

Aggregate acceptance rates hide too much. The diversity and strategy diagnostics show whether the benchmark is balanced, which Cypher operators it exercises, and where residual concentration remains. Figure 12 shows the useful diversity picture: category, graph-category, and difficulty balance are strong by construction, while schema and value coverage remain quantities to monitor during refresh.

Metric	Value
PIPE-Diversity index	0.549
Question Distinct-1	0.039
Question Distinct-2	0.085
Question adjusted Distinct-2	0.266
Question self-BLEU-2 (sampled)	0.813
Mean nearest-neighbor question Jaccard	0.775
Unique query-signature ratio	0.041
Top query-signature share	0.083
Template-family entropy	0.826
Operator-combination entropy	0.880
Unique structural substructures	134
Category normalized entropy	1.000
Graph-category normalized entropy	0.980
Difficulty normalized entropy	0.998
Label coverage	0.941
Relationship-type coverage	0.708
Property-name coverage	0.426
Unique grounded-value ratio	0.357
Grounded values exactly quoted	0.823
Aggregation / negation / ordering rates	0.500 / 0.125 / 0.125

Table 22: Diversity diagnostics for the full exported benchmark. PIPE-Diversity is a geometric mean of lexical, query-template, structural, schema, value, and balance components; component rows are shown so the composite score does not hide residual concentration.

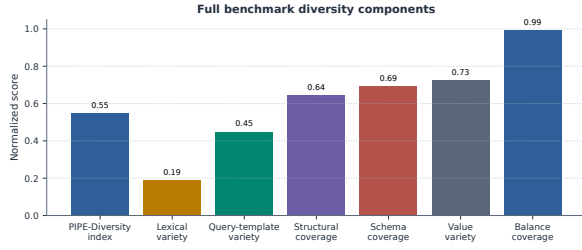


Figure 12: Diversity diagnostics for the full 3,000-example export. Category, graph-category, and difficulty balance are strong by construction; schema coverage and query-signature diversity expose remaining concentration from the seeded-template run.

F.2 Diversity-Governed Selection

Table 23 reports the first diversity-improvement pass. We select a target-50-per-graph/category subset from the 3,000 accepted examples and compare it with a hash-balanced random subset of the same size. The selector uses an MMR-style novelty score over query signatures, template families, structural substructures, schema atoms, entity values, and question tokens after all quality gates have passed. It improves the PIPE-Diversity index, unique query-signature ratio, unique structural substructures, adjusted Distinct-2, and property coverage while preserving a signature-disjoint 640/80/80 split. The mixed template-family entropy and self-BLEU result is useful rather than embarrassing: it shows that post-hoc selection can improve coverage, but stronger diversity requires oversampling or inducing more source templates during generation when a graph/category cell has only one or two viable signatures.

Metric	Random balanced	Diversity governed	Δ
PIPE-Diversity index	0.557	0.575	+0.017
Unique query-signature ratio	0.062	0.135	+0.073
Top signature share (lower better)	0.062	0.062	+0.000
Template-family entropy	0.903	0.868	-0.035
Operator-combination entropy	0.901	0.911	+0.010
Unique structural substructures	97.000	134.000	+37.000
Question self-BLEU-2 (lower better)	0.850	0.866	+0.016
Adjusted Distinct-2	0.266	0.284	+0.018
Property coverage	0.407	0.426	+0.019

Table 23: Balanced subset comparison at the same graph/category target. The diversity-governed selector applies MMR-style novelty over Cypher signatures, template families, structural substructures, schema atoms, values, and question tokens after quality gates have already passed; structural/schema gains are reported alongside residual template concentration.

F.3 Cypher Strategy Coverage

Strategy diagnostics complement category balance. Two questions can share a category while exercising different Cypher behavior. Conversely,

a category-balanced dataset can still miss joins, paths, negation, ranking, or bounded-result patterns. The strategy matrix and downstream strategy outcomes show what the benchmark actually exercises.

Primary strategy	Examples	Share	Rel. patterns	Return arity	Downstream exec. acc.
Aggregation	1125	0.375	1.42	1.00	0.396
Join-heavy	375	0.125	2.33	2.33	0.000
Negation	375	0.125	1.89	2.84	0.000
Order/rank	375	0.125	1.87	3.41	0.000
Path	375	0.125	1.37	3.00	0.000
Single hop	375	0.125	1.00	2.33	0.324

Table 24: Cypher strategy diagnostics over the full 3,000-example export. Strategy tags are derived from generated Cypher structure rather than from category labels; downstream execution accuracy is reported on the full held-out test split when a strategy appears there.

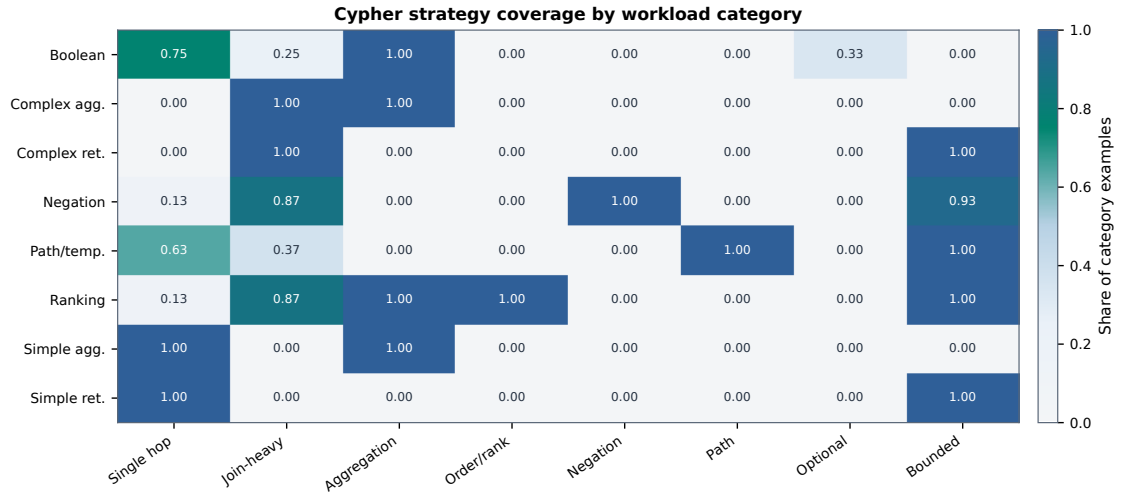


Figure 13: Cypher strategy coverage by workload category. Category balancing does not by itself guarantee operator coverage; the strategy matrix shows where the benchmark exercises single-hop retrieval, joins, aggregation, ordering, negation, path patterns, optional matches, and bounded-result queries.

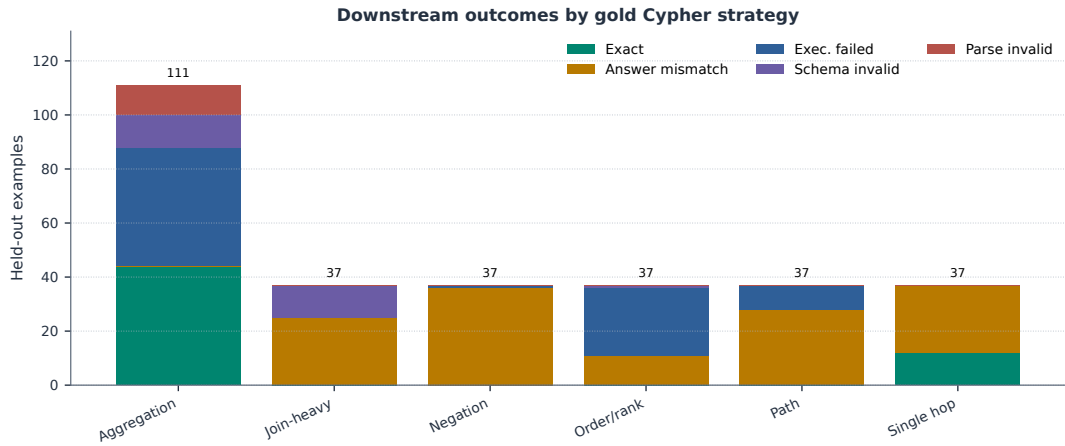


Figure 14: Downstream outcomes by gold Cypher strategy on the full held-out test split. The local Qwen3.5-9B baseline fails differently across strategies: aggregation mixes exact matches with execution failures, while join-heavy, negation, path, and ranking examples are dominated by semantically wrong executable queries or execution failures.

G Supplementary Evaluator Metrics

Table 25 lists optional text-overlap metrics for debugging answer rendering and near-match behavior. These are not correctness metrics; the reported correctness results use execution accuracy and answer-set F1.

Metric	Supported use and primary citation
ROUGE-1/2/L	Reference overlap over serialized answer sets and query strings (Lin, 2004)
BLEU	Sentence-level reference overlap with brevity penalty (Papineni et al., 2002)
METEOR	Exact-token precision/recall with fragmentation penalty (Banerjee and Lavie, 2005)
BERTScore	Optional contextual-embedding similarity (Zhang et al., 2020)
FrugalScore	Optional efficient learned NLG metric (Kamal Eddine et al., 2022)
Cosine	Token-count vector cosine similarity (Manning et al., 2008)
Jaro-Winkler	Character-level near-match similarity (Jaro, 1989; Winkler, 1990)
Exact match	Raw and normalized string exact match (Rajpurkar et al., 2016)

Table 25: Supplementary reference-based text metrics supported by the PIPE-Cypher evaluator. These metrics are useful for debugging answer rendering, paraphrase sensitivity, and near-match behavior, but they do not replace execution accuracy or answer-set F1 for Text2Cypher correctness. BERTScore and FrugalScore are optional integrations because they require additional metric/model packages.

H Governance, Judge Calibration, and Deployment Details

The deployment details below matter because enterprise failures rarely come from a single model call. They come from missing safety boundaries, unclear value policies, weak audit trails, or no explanation for why an example was accepted. The validator cascade, prompt contracts, automation comparison, and judge audit describe the controls that make PIPE-Cypher a governed local workflow rather than a manual dataset-writing exercise.

H.1 Validation Cascade

Table 26 lists the deterministic and judge gates in execution order. It shows how an enterprise deployment keeps unsafe, schema-invalid, empty, or semantically weak examples out of exported benchmarks.

Gate or ledger	Count	Denominator
Export accepted	3,000	3,000
Read-only safety	3,000	3,000
Syntax validity	3,000	3,000
Schema/value validity	3,000	3,000
Execution success	3,000	3,000
LLM judge pass	3,000	3,000
Rejected candidates logged	1,925	4,925

Table 26: PIPE-Cypher validation cascade for the full export and its logged candidate ledger. Unlike Mind the Query, human review is calibration-only.

H.2 Rewrite and Governance Audits

Table 27 addresses rewrite preservation directly. In the reported generation records, generated Cypher was already identical to normalized Cypher, so the accepted benchmark does not rely on a semantics-changing RETURN DISTINCT insertion or projection rewrite. Rewriting is still part of the pipeline, but in this run its role is conservative validation and logging rather than silent semantic alteration.

Rewrite audit property	Value
Generation records audited	4,925
Accepted records audited	3,000
Records changed by normalization	0
Accepted records changed	0
RETURN DISTINCT insertions	0
Accepted RETURN DISTINCT insertions	0
Rewrite-skip reasons logged	196
Live comparisons required	0
Answer-set equality in comparisons	0

Table 27: Rewrite prevalence and impact audit over reported generation records. When no generated query differs from its normalized form, no live original/normalized re-execution is required for semantic drift.

Table 28 separates direction, schema/value, syntax/parser, and read-only issues across generation, ablation, and downstream prediction artifacts. The full generation records have no direction failures after validation. Downstream predictions still contain direction errors, which is exactly the kind of graph-specific failure a Cypher benchmark should expose.

Evidence source	Direction	Schema/value	Syntax/parser	Read-only
Full generation records	0	2	0	0
Target-size ablations	260	988	5	0
Downstream predictions	25	9	12	0
Combined	285	999	17	0

Table 28: Governance failure audit. Direction errors, schema/value errors, syntax/parser failures, and read-only violations are counted separately so the appendix shows which Cypher-specific gates do real work.

H.3 Gate-Impact Counterfactual

Table 29 summarizes the first gate that blocks each non-accepted candidate. This gives the counterfactual view missing from a yield-only ablation: it shows what kind of bad example would enter the benchmark if a deployment weakened duplicate/diversity controls, non-empty execution, judge review, schema validation, direction checking, or read-only safety.

First blocking gate	Candidates	Share
Accepted	3,000	0.609
Duplicate/diversity	1,366	0.277
Empty result	486	0.099
Judge reject	67	0.014
Schema	2	0.000
Execution failure	4	0.001

Table 29: Counterfactual first-blocking-gate audit over generation records. The table shows which failure class would enter the benchmark if that gate were removed or weakened.

H.4 Privacy and Redaction Audit

Table 30 evaluates the redaction policy rather than merely describing it. The audit builds a sensitive-value set from entity bindings, quoted Cypher literals, reverse-grounding literals, and string-valued execution samples. It then applies the configured redactor and exact-matches the raw values against the redacted question, Cypher, entity, and result fields. This does not replace a tenant’s PII classifier, but it gives reviewers a measurable privacy check for the value-bearing fields PIPE-Cypher itself creates.

Redaction audit property	Value
Examples audited	3,000
Sensitive values checked	10,956
Examples with sensitive values	2,970
Examples with residual raw values	0
Residual raw-value matches	0
Residual rate per checked value	0.000
Unique placeholders	3,754
Reused placeholders	2,342
Max placeholder frequency	337

Table 30: Exact-match redaction audit over value-bearing benchmark surfaces. The audit checks entity bindings, quoted Cypher literals, reverse grounding literals, and string-valued result samples after applying the configured redaction policy.

H.5 Operational Accounting

Table 31 reports local-run accounting from completed generation records. These numbers are for deployment planning: acceptance rate and graph execution latency explain throughput bottlenecks without turning the analysis into a paid-API cost comparison.

Scope	Records	Accepted	Acceptance	Exec. p50 ms	Exec. p95 ms
Overall	4,925	3,000	0.609	11.995	32.832
FinBench	3,405	2,000	0.587	11.837	32.398
SNB	1,520	1,000	0.658	12.420	38.558

Table 31: Operational accounting from completed generation records. These are local-run latency and acceptance diagnostics, not paid-API cost claims.

H.6 Prompt Profiles and Contracts

Table 32 documents the prompt-profile factors used for ablation planning. The full prompt contracts follow immediately after the table, including the exact LLM-judge system prompt and user prompt template used for reported judge decisions.

Profile	Added constraint	Intended evidence
Schema-only	Use only visible schema.	Baseline for schema-grounded prompting.
Instructions	Exact values, datatype rules, no nested aggregations.	Targets MTQ-style prompt refinements.
Examples	Placeholderized retrieved NL-Cypher pairs.	Tests whether examples help without extra governance.
Examples + instructions	Few-shot plus explicit rules.	Closest controlled analogue to MTQ Table 5.
Full governed	Production-derived Cypher hints, AST-safe rewrites, judge gate.	PIPE-Cypher production setting.

Table 32: Prompt profiles implemented for Mind-the-Query-style prompt-factorial evaluation. Results are reported only for completed, audited target-50-or-larger suites.

I Prompt Contracts

PIPE-Cypher treats prompts as versioned implementation artifacts. The list summarizes the prompt contracts used for generation, repair, judging, and

downstream evaluation; hashes fingerprint the full prompt constants in the codebase.

1. **Template generation.** Stage: Workload proposal. SHA-256: 10b25b.

Contract. Schema-only labels, relationships, properties, and categorical values; Realistic enterprise analyst wording; At most two typed slots and JSON-only output

2. **Reverse binding.** Stage: Graph grounding. SHA-256: 48e949.

Contract. Read-only MATCH/WHERE/RETURN DISTINCT/LIMIT only; Slot variables named exactly as requested; Forward relationship directions from the schema

3. **Cypher generation.** Stage: Candidate query. SHA-256: 61c557.

Contract. Only schema-visible constructs and observed directions; RETURN DISTINCT for set returns and exact equality for quoted values; Context columns, categorical hints, placeholderized retrieval, and no writes

4. **Repair.** Stage: Validation feedback. SHA-256: 56c419.

Contract. Preserve question intent while fixing validation or execution issues; Keep query read-only and schema-grounded; Return only corrected Cypher

5. **LLM judge.** Stage: Quality gate. SHA-256: 421c7b.

Contract. Inputs include question, Cypher, relevant schema excerpt, execution rows, and validation summary; Strict JSON scores for ambiguity, semantic alignment, schema use, and difficulty; Categorical values constrain query literals, not observed result-row values; Pass only useful, unambiguous enterprise benchmark examples

6. **Downstream Text2Cypher.** Stage: Model evaluation. SHA-256: 4c07ff.

Contract. Read-only Cypher only; Schema-visible constructs and exact direction preservation; RETURN DISTINCT, count/ranking rules, and no explanations

I.1 LLM Judge Prompt Used in Reported Runs

The judge runs only after deterministic validation and live execution. For each reviewed example, we fill the template below with the candidate question, Cypher, relevant schema excerpt, sampled execution rows, and validation summary. This is the prompt used for all reported LLM-judge decisions.

System prompt:

You are an expert benchmark engineer. Return strict JSON only. No markdown or extra text.

User prompt template:

You are judging whether an NL-to-Cypher benchmark example is acceptable for an enterprise benchmark.

Graph schema:

{schema}

Question:

{question}

Cypher:

{cypher}

Execution sample:

{rows}

Validation summary:

{validation}

Return strict JSON with:

- pass: boolean
- ambiguity_score: number from 0 to 1, lower is better
- semantic_alignment_score: number from 0 to 1
- schema_use_score: number from 0 to 1
- difficulty: one of easy, medium, hard
- failure_reason: short string, empty if pass is true

Pass only if the question is unambiguous, the Cypher answers it, the schema use is valid, and the result would be useful in an enterprise benchmark.

- Categorical property values in the schema constrain literal values written in the Cypher query.
- Do not reject because the execution sample returns a value that is absent from the categorical-value list; result rows are observed graph outputs.
- If deterministic validation says schema use is valid, lower schema_use_score only when the Cypher itself uses nonexistent schema elements, invalid relationship directions, or invalid literal values.

I.2 Automation and Calibration

Table 33 gives the deployment contrast with Mind the Query, and Table 34 reports the completed human calibration packet. Human review has not disappeared from the research process. It has moved from a production gate to post-hoc calibration evidence.

Human annotation protocol. One external annotator labeled the frozen 80-row audit packet after generation was complete. The packet sampled judge-accepted and judge-rejected candidates across FinBench/SNB categories. For each

row, the annotator inspected the NL question, Cypher, graph/category metadata, execution evidence, and judge decision, then filled a binary `human_accept` label and optional notes under the rubric in Appendix I: accept only if the question is clear and the Cypher is read-only, schema-grounded, directionally plausible, exact about quoted values, and semantically aligned with the question. The annotator’s labels calibrate the local judge and are reported only in aggregate. We do not report protected demographic attributes because a single annotator would be identifiable; recruitment and compensation details are recorded in the Responsible NLP checklist. The annotation protocol was determined exempt by an IRB.

Dimension	Mind the Query	PIPE-Cypher
Generation review gate	Manual logical review	Deterministic gates + local LLM judge
Human effort	Reported 1,400 person-hours	80-row post-hoc judge calibration audit
Private values	Public benchmark values	Configurable sampling and redacted export
Refresh	Static dataset release	Rerunnable private benchmark factory
Model endpoint	Gemini in their reported pipeline	Local Qwen3.5-9B endpoint

Table 33: Industry deployment contrast with Mind the Query. PIPE-Cypher focuses on private refreshable benchmark generation rather than a one-time public dataset.

Audit packet property	Value
Rows	80
Human annotators	1 external annotator
IRB status	exempt determination
Use of human labels	post-hoc calibration only
Judge accept / reject	40 / 40
FinBench / SNB rows	48 / 32
Easy / medium rows	26 / 54
Labeled rows	80
Calibration status	complete
Judge-human agreement / κ	0.800 / 0.600
Judge precision (95% CI)	1.000 (0.912–1.000)
Judge recall (95% CI)	0.714 (0.585–0.816)
False-accept rate (95% CI)	0.000 (0.000–0.138)
False-reject rate (95% CI)	0.286 (0.184–0.415)

Table 34: Post-hoc judge calibration packet coverage and judge-human agreement. Human labels calibrate the automated gate after generation and are not used as a generation gate.

J Representative Accepted Examples

The examples below are selected in stable identifier order from tracked evidence snapshots, one per graph/category cell when available. They show the NL question, accepted Cypher, structural tags, gate status, and a bounded execution-result sample. ICIJ examples are redacted before rendering, preserving query structure while removing value-bearing strings.

1. FinBench / Boolean Existence / easy. Question: Does account '187743809466009406' have any outgoing transfer?

```
MATCH (src:Account {accountId:
'187743809466009406'})
-[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT COUNT(DISTINCT src)
> 0
AS HasOutgoingTransfer
```

Structure: single_hop, aggregation.

Relationships: TRANSFER_TO.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {HasOutgoingTransfer: True}; observed rows: 1

2. FinBench / Complex Aggregation / medium. Question: What is the total transferred amount from accounts owned by person 'Gwar'?

```
MATCH (p:Person {personName:
'Gwar'})
-[:OWN_ACCOUNT]->
(src:Account)-[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT SUM(t.amount) AS
TotalTransferredAmount
```

Structure: join_heavy, aggregation.

Relationships: OWN_ACCOUNT, TRANSFER_TO.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {TotalTransferredAmount: 14972318.41}; observed rows: 1

3. FinBench / Complex Retrieval / easy. Question: Which accounts received transfers from accounts owned by person 'Zof'?

```
MATCH (p:Person {personName:
'Zof'})
-[:OWN_ACCOUNT]-> (src:Account)
-[:TRANSFER_TO]-> (dst:Account)
RETURN DISTINCT dst.accountId AS
AccountId, dst.accountType AS
AccountType, dst.isBlocked AS
IsBlocked
LIMIT 300
```

Structure: join_heavy, bounded_result.

Relationships: OWN_ACCOUNT, TRANSFER_TO.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {AccountId: 4687402787162554854, AccountType: credit card, IsBlocked: False}; observed rows: 3

4. FinBench / Negation Difference / medium. Question: Which accounts owned by person 'Kant' have not sent any transfers?

```
MATCH (p:Person {personName:
'Kant'})
-[:OWN_ACCOUNT]-> (a:Account)
WHERE NOT (a) -[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT a.accountId AS
AccountId, a.accountType AS
AccountType,
a.isBlocked AS IsBlocked
LIMIT 300
```

Structure: join_heavy, negation, bounded_result.

Relationships: OWN_ACCOUNT, TRANSFER_TO.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {AccountId: 4739757132830738341, AccountType: merchant account, IsBlocked: False}; observed rows: 1

5. FinBench / Path Temporal / medium. Question: Which accounts can receive money within two transfer hops from accounts owned by person 'Sossamon'?

```
MATCH (p:Person {personName:
'Sossamon'})
-[:OWN_ACCOUNT]->
(src:Account) -[:TRANSFER_TO*1..2]->
(dst:Account)
RETURN DISTINCT dst.accountId AS
AccountId, dst.accountType AS
AccountType, dst.isBlocked AS
IsBlocked
LIMIT 300
```

Structure: join_heavy, path, bounded_result.

Relationships: OWN_ACCOUNT, TRANSFER_TO.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {AccountId: 4687402787162554854, AccountType: credit card, IsBlocked: False}; observed rows: 4

6. FinBench / Ranking Topk / medium. Question: For accounts owned by person 'Barry', which account sent the highest total transfer amount?

1265	MATCH (p:Person {personName:	9. ICIJ Offshore Leaks / Boolean Existence	1319
1266	'Barry'})	/ medium. Question: Does offshore entity	1320
1267	-[:OWN_ACCOUNT]->	'ENTITY_VALUE_1' have a registered ad-	1321
1268	(src:Account)-[:TRANSFER_TO]->	dress?	1322
1269	(:Account)		
1270	WITH src, SUM(t.amount) AS		
1271	totalAmount		
1272	RETURN DISTINCT src.accountId AS	MATCH (e:Entity {name:	1323
1273	AccountId, src.accountType AS	'ENTITY_VALUE_1'})	1324
1274	AccountType, src.isBlocked AS	OPTIONAL MATCH (e)	1325
1275	IsBlocked,	-[:registered_address]->	1326
1276	totalAmount	(addr:Address)	1327
1277	ORDER BY totalAmount DESC	RETURN DISTINCT COUNT(addr) > 0 AS	1328
1278	LIMIT 1	HasRegisteredAddress	1329
1279	Structure: join_heavy, aggregation, or-	Structure: single_hop, aggregation, optional.	1330
1280	der_rank, bounded_result.	Relationships: registered_address.	1331
1281	Relationships: OWN_ACCOUNT, TRANS-	Gates: RO/Syn/Schema/Exec/Judge.	1332
1282	FER_TO.	Result sample: {HasRegisteredAddress:	1333
1283	Gates: RO/Syn/Schema/Exec/Judge.	True}; observed rows: 1	1334
1284	Result sample: {AccountId:	10. ICIJ Offshore Leaks / Complex Aggrega-	1335
1285	4732438783436260772, AccountType:	tion / easy. Question: How many distinct	1336
1286	internet account, IsBlocked: False}; observed	officers are connected to entities in jurisdic-	1337
1287	rows: 1	tion 'JURISDICTION_VALUE_1'?	1338
1288	7. FinBench / Simple Aggregation / easy. Ques-		
1289	tion: How many accounts are owned by per-	MATCH (o:Officer) -[:officer_of]->	1339
1290	son 'Kaewsuktae'?	(e:Entity {jurisdiction:	1340
1291		'JURISDICTION_VALUE_1'})	1341
1292	MATCH (p:Person {personName:	RETURN DISTINCT COUNT(DISTINCT o) AS	1342
1293	'Kaewsuktae'}) -[:OWN_ACCOUNT]->	OfficerCount	1343
1294	(a:Account)	Structure: single_hop, aggregation.	1344
1295	RETURN DISTINCT COUNT(DISTINCT a) AS	Relationships: officer_of.	1345
1296	AccountCount	Gates: RO/Syn/Schema/Exec/Judge.	1346
1297	Structure: single_hop, aggregation.	Result sample: {OfficerCount: 2}; observed	1347
1298	Relationships: OWN_ACCOUNT.	rows: 1	1348
1299	Gates: RO/Syn/Schema/Exec/Judge.	11. ICIJ Offshore Leaks / Complex Retrieval	1349
1300	Result sample: {AccountCount: 1}; observed	/ easy. Question: Which officers share a	1350
1301	rows: 1	registered address with offshore entity 'EN-	1351
1302	8. FinBench / Simple Retrieval / easy. Ques-	TITY_VALUE_1'?	1352
1303	tion: Which accounts are owned by person		
1304	'Barry'?	MATCH (e:Entity {name:	1353
1305		'ENTITY_VALUE_1'})	1354
1306	MATCH (p:Person {personName:	-[:registered_address]->	1355
1307	'Barry'})	(addr:Address)	1356
1308	-[:OWN_ACCOUNT]-> (a:Account)	<-[:registered_address]- (o:Officer)	1357
1309	RETURN DISTINCT a.accountId AS	RETURN DISTINCT o.node_id AS	1358
1310	AccountId, a.accountType AS	OfficerId,	1359
1311	AccountType,	o.name AS OfficerName, addr.address	1360
1312	a.isBlocked AS IsBlocked	AS	1361
1313	LIMIT 300	RegisteredAddress	1362
1314	Structure: single_hop, bounded_result.	LIMIT 300	1363
1315	Relationships: OWN_ACCOUNT.	Structure: join_heavy, bounded_result.	1364
1316	Gates: RO/Syn/Schema/Exec/Judge.	Relationships: registered_address.	1365
1317	Result sample: {AccountId:	Gates: RO/Syn/Schema/Exec/Judge.	1366
1318	4732438783436260772, AccountType:	Result sample: {OfficerId: OFFICER_ID_1,	1367
	internet account, IsBlocked: False}; observed	OfficerName: OFFICER_NAME_1, Regis-	1368
	rows: 1	teredAddress: ADDRESS_1}; observed rows:	1369
		8	1370

1371 12. **ICIJ Offshore Leaks / Negation Difference**
 1372 / **easy.** *Question:* Which offshore entities in
 1373 jurisdiction 'JURISDICTION_VALUE_1' do
 1374 not have a registered address?

```
1375 MATCH (e:Entity {jurisdiction:
1376 'JURISDICTION_VALUE_1'})
1377 WHERE NOT (e)
1378 -[:registered_address]->
1379 (:Address)
1380 RETURN DISTINCT e.node_id AS
1381 EntityId,
1382 e.name AS EntityName, e.jurisdiction
1383 AS
1384 Jurisdiction
1385 LIMIT 300
```

1386 *Structure:* single_hop, negation,
 1387 bounded_result.

1388 *Relationships:* registered_address.

1389 *Gates:* RO/Syn/Schema/Exec/Judge.

1390 *Result sample:* {EntityId: ENTITY_ID_1, En-
 1391 tityName: ENTITY_NAME_1, Jurisdiction:
 1392 JURISDICTION_1}; observed rows: 2

1393 13. **ICIJ Offshore Leaks / Path Temporal**
 1394 / **medium.** *Question:* Which officers
 1395 share offshore entities with officer 'OFFI-
 1396 CER_VALUE_1', and when did each connec-
 1397 tion start?

```
1398 MATCH
1399 (src:Officer)-[srcRel:officer_of]->
1400 (entity:Entity)
1401 <-[dstRel:officer_of]-
1402 (dst:Officer)
1403 WHERE trim(src.name) =
1404 'OFFICER_VALUE_1'
1405 AND dst <> src
1406 RETURN DISTINCT dst.node_id AS
1407 OfficerId, dst.name AS OfficerName,
1408 entity.name AS SharedEntityName,
1409 dstRel.start_date AS
1410 ConnectionStartDate
1411 LIMIT 300
```

1412 *Structure:* join_heavy, negation,
 1413 bounded_result.

1414 *Relationships:* officer_of.

1415 *Gates:* RO/Syn/Schema/Exec/Judge.

1416 *Result sample:* {ConnectionStartDate:
 1417 DATE_1, OfficerId: OFFICER_ID_1, Offi-
 1418 cerName: OFFICER_NAME_1}; observed
 1419 rows: 15

1420 14. **ICIJ Offshore Leaks / Ranking Topk /**
 1421 **medium.** *Question:* Which jurisdictions have
 1422 the most offshore entities?

```
1423 MATCH (e:Entity)
1424 WHERE e.jurisdiction IS NOT NULL
1425 WITH e.jurisdiction AS jurisdiction,
1426 COUNT(DISTINCT e) AS entityCount
1427 RETURN DISTINCT jurisdiction,
1428 entityCount
1429 ORDER BY entityCount DESC
1430 LIMIT 10
```

Structure: node_scan, aggregation, or- 1431
 der_rank, negation, bounded_result. 1432

Relationships: none. 1433

Gates: RO/Syn/Schema/Exec/Judge. 1434

Result sample: {entityCount: 209634, juris- 1435
 diction: JURISDICTION_1}; observed rows: 1436
 10 1437

1438 15. **ICIJ Offshore Leaks / Simple Aggrega-**
 1439 **tion / easy.** *Question:* How many off- 1439
 shore entities are connected to officer 'OF- 1440
 FICER_VALUE_1'? 1441

```
1442 MATCH (o:Officer) -[:officer_of]->
1443 (e:Entity)
1444 WHERE trim(o.name) =
1445 'OFFICER_VALUE_1'
1446 RETURN DISTINCT COUNT(DISTINCT e) AS
1447 OffshoreEntityCount
```

Structure: single_hop, aggregation. 1448

Relationships: officer_of. 1449

Gates: RO/Syn/Schema/Exec/Judge. 1450

Result sample: {OffshoreEntityCount: 1}; ob- 1451
 served rows: 1 1452

1453 16. **ICIJ Offshore Leaks / Simple Retrieval /**
 1454 **easy.** *Question:* Which offshore entities is 1454
 officer 'OFFICER_VALUE_1' connected to? 1455

```
1456 MATCH (o:Officer)-[r:officer_of]->
1457 (e:Entity)
1458 WHERE trim(o.name) =
1459 'OFFICER_VALUE_1'
1460 RETURN DISTINCT e.node_id AS
1461 EntityId,
1462 e.name AS EntityName, e.jurisdiction
1463 AS
1464 Jurisdiction, r.link AS Link
1465 LIMIT 300
```

Structure: single_hop, bounded_result. 1466

Relationships: officer_of. 1467

Gates: RO/Syn/Schema/Exec/Judge. 1468

Result sample: {EntityId: ENTITY_ID_1, En- 1469
 tityName: ENTITY_NAME_1, Jurisdiction: 1470
 JURISDICTION_1}; observed rows: 1 1471

1472 17. **SNB / Boolean Existence / medium. Ques-**
 1473 **tion:** Does person with id 6597069766828 1473
 like any post? 1474

```
1475 MATCH (p:Person {id:
1476 6597069766828})
1477 OPTIONAL MATCH (p) -[:LIKES]->
1478 (post:Post)
1479 RETURN DISTINCT COUNT(post) > 0 AS
1480 LikesAnyPost
```

Structure: single_hop, aggregation, optional. 1481

Relationships: LIKES. 1482

Gates: RO/Syn/Schema/Exec/Judge. 1483

1484	<i>Result sample:</i> {LikesAnyPost: True}; ob-	MATCH (src:Person {id:	1537
1485	erved rows: 1	4398046511136})	1538
		-[:KNOWS*1..2]-> (dst:Person)	1539
1486	18. SNB / Complex Aggregation / medium.	RETURN DISTINCT dst.id AS PersonId,	1540
1487	<i>Question:</i> How many distinct posts	dst.firstName AS FirstName,	1541
1488	are in forums joined by person with id	dst.lastName	1542
1489	6597069766845?	AS LastName	1543
		LIMIT 200	1544
1490			
1491	<pre>MATCH (forum:Forum) -[:HAS_MEMBER]-></pre>		
1492	<pre>(p:Person {id: 6597069766845})</pre>		
1493	<pre>MATCH (forum) -[:CONTAINER_OF]-></pre>		
1494	<pre>(post:Post)</pre>		
1495	<pre>RETURN DISTINCT COUNT(DISTINCT post)</pre>		
1496	<pre>AS</pre>		
	<pre>JoinedForumPostCount</pre>		
1497	<i>Structure:</i> join_heavy, aggregation.	<i>Structure:</i> single_hop, path, bounded_result.	1545
1498	<i>Relationships:</i> CONTAINER_OF,	<i>Relationships:</i> KNOWS.	1546
1499	HAS_MEMBER.	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	1547
1500	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	<i>Result sample:</i> {FirstName: Rafael,	1548
1501	<i>Result sample:</i> {JoinedForumPostCount: 1};	LastName: Fernández, PersonId:	1549
1502	observed rows: 1	4398046511333}; observed rows: 25	1550
1503	19. SNB / Complex Retrieval / easy. Question:	22. SNB / Ranking Topk / medium. Ques-	1551
1504	Which people are members of forums contain-	<i>tion:</i> Which city records are linked	1552
1505	ing posts tagged 'Manuel_Noriega'?	from the most organisation records through	1553
		:IS_LOCATED_IN?	1554
1506	<pre>MATCH (forum:Forum) -[:HAS_MEMBER]-></pre>	<pre>MATCH (s:Organisation)</pre>	1555
1507	<pre>(p:Person), (forum)</pre>	<pre>-[:IS_LOCATED_IN]-> (e:City)</pre>	1556
1508	<pre>-[:CONTAINER_OF]-></pre>	WITH e, COUNT(DISTINCT s) AS	1557
1509	<pre>(post:Post) -[:HAS_TAG]-> (tag:Tag</pre>	relatedCount	1558
1510	<pre>{name: 'Manuel_Noriega'})</pre>	RETURN DISTINCT e.id AS TargetId,	1559
1511	<pre>RETURN DISTINCT p.id AS PersonId</pre>	e.name	1560
1512	<pre>LIMIT 200</pre>	AS TargetName, relatedCount	1561
		ORDER BY relatedCount DESC	1562
1513	<i>Structure:</i> join_heavy, bounded_result.	LIMIT 10	1563
1514	<i>Relationships:</i> CONTAINER_OF,	<i>Structure:</i> single_hop, aggregation, or-	1564
1515	HAS_MEMBER, HAS_TAG.	der_rank, bounded_result.	1565
1516	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	<i>Relationships:</i> IS_LOCATED_IN.	1566
1517	<i>Result sample:</i> {PersonId: 4398046511220};	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	1567
1518	observed rows: 19	<i>Result sample:</i> {TargetId: 164, TargetName:	1568
1519	20. SNB / Negation Difference / easy. Question:	Kolkata, relatedCount: 130}; observed rows:	1569
1520	Which person records are not linked from any	10	1570
1521	message record through :HAS_CREATOR?	23. SNB / Simple Aggregation / easy. Question:	1571
1522	<pre>MATCH (p:Person)</pre>	How many posts are tagged 'Vietnam'?	1572
1523	<pre>WHERE NOT EXISTS((:Message)</pre>	<pre>MATCH (post:Post) -[:HAS_TAG]-></pre>	1573
1524	<pre>-[:HAS_CREATOR]-> (p))</pre>	(tag:Tag	1574
1525	<pre>RETURN DISTINCT p.id AS PersonId,</pre>	{name: 'Vietnam'})	1575
1526	<pre>p.firstName AS PersonFirstName,</pre>	RETURN DISTINCT COUNT(DISTINCT post)	1576
1527	<pre>p.lastName AS PersonLastName</pre>	AS	1577
		PostCount	1578
1528	<i>Structure:</i> single_hop, negation.	<i>Structure:</i> single_hop, aggregation.	1579
1529	<i>Relationships:</i> HAS_CREATOR.	<i>Relationships:</i> HAS_TAG.	1580
1530	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	1581
1531	<i>Result sample:</i> {PersonFirstName: R., Per-	<i>Result sample:</i> {PostCount: 1}; observed	1582
1532	sonId: 8796093022313, PersonLastName:	rows: 1	1583
1533	Rao}; observed rows: 25	24. SNB / Simple Retrieval / easy. Ques-	1584
1534	21. SNB / Path Temporal / easy. Question:	<i>tion:</i> Which post IDs did person with id	1585
1535	Which people are within two knows hops of	4398046511124 like?	1586
1536	person with id 4398046511136?	<pre>MATCH (p:Person {id:</pre>	1587
		4398046511124})	1588
		-[:LIKES]-> (post:Post)	1589
		RETURN DISTINCT post.id AS PostId	1590
		LIMIT 200	1591

1592 *Structure:* single_hop, bounded_result.
1593 *Relationships:* LIKES.
1594 *Gates:* RO/Syn/Schema/Exec/Judge.
1595 *Result sample:* {PostId: 343597385744}; ob-
1596 served rows: 5